

Bootstrap Classes Used in the Portfolio

Bootstrap 5.3.8 reference for the classes actually used in the supplied HTML. Custom CSS classes such as hero-section and logo are excluded.

1. Quick Reference Table

Class	Category	Use Case	Example + Other Options
container-fluid	Layout	Full-width responsive container.	<code><div class="container-fluid">Content</div></code> Options: container — centered max-width container.
py-3	Spacing	Adds padding to the top and bottom.	<code><div class="py-3">Header</div></code> Options: py-1, py-2, py-4, py-5; p-3 adds padding on all sides.
row	Grid	Creates a Bootstrap grid row and flex container for columns.	<code><div class="row">...</div></code> Options: row-cols-2, row-cols-3, row-cols-4 can control automatic column counts.
align-items-center	Flexbox	Vertically centers flex items on the cross axis.	<code><div class="row align-items-center">...</div></code> Options: align-items-start, align-items-end, align-items-stretch.
col-md-6	Grid	Uses 6 of 12 columns from the md breakpoint upward.	<code><div class="col-md-6">Half</div></code> Options: col-md-3 = 25%; col-md-4 = 33.33%; col-md-8 = 66.67%; col-md-12 = 100%.
text-md-end	Text / Alignment	Aligns text to the end from md and above.	<code><div class="text-md-end">...</div></code> Options: text-start, text-center, text-end; responsive versions can use sm, lg, xl, xxl.
me-3	Spacing	Adds margin to the end side; normally right margin in LTR layouts.	<code>Home</code> Options: me-1, me-2, me-4, me-5; ms-3 applies margin to the start side.
h-100	Sizing	Sets height to 100% of the parent when the parent has a usable height.	<code><div class="h-100">...</div></code> Options: h-25, h-50, h-75, h-auto.
align-items-start	Flexbox	Aligns flex items to the start of the cross axis; in a row this is normally the top.	<code><div class="row align-items-start">...</div></code> Options: align-items-center, align-items-end, align-items-stretch.
col-md-5	Grid	Uses 5 of 12 columns from md upward.	<code><div class="col-md-5">...</div></code> Options: col-md-4, col-md-6, col-md-7, col-md-8, etc.
col-md-7	Grid	Uses 7 of 12 columns from md upward.	<code><div class="col-md-7">...</div></code> Options: col-md-5 pairs with it to make 12 columns.
pt-5	Spacing	Adds large padding to the top.	<code><div class="pt-5">Hero content</div></code> Options: pt-1 through pt-5; py-5 adds top and bottom padding.
container	Layout	Responsive centered container with breakpoint-based max-width.	<code><div class="container">Programs</div></code> Options: container-fluid; container-sm, md, lg, xl, xxl.
mt-4	Spacing	Adds margin to the top.	<code><div class="mt-4">Programs</div></code> Options: mt-1, mt-2, mt-3, mt-5; mb-4 for bottom margin; my-4 for top and bottom.
text-center	Text / Alignment	Centers inline text and inline content.	<code><div class="text-center">Program</div></code> Options: text-start, text-end.

Class	Category	Use Case	Example + Other Options
g-3	Grid / Gutters	Adds horizontal and vertical grid gutters.	<div class="row g-3">...</div> Options: g-0 through g-5; gx-3 horizontal only; gy-3 vertical only.
col-md-3	Grid	Uses 3 of 12 columns from md upward, giving four equal columns.	<div class="col-md-3">Program 1</div> Options: col-md-6 gives two columns; col-md-4 gives three; col-md-12 gives one.
border	Borders	Adds a border around the element.	<div class="border">Box</div> Options: border-0 removes it; border-top, border-end, border-bottom, border-start.
p-4	Spacing	Adds padding on all four sides.	<div class="p-4">Program 1</div> Options: p-0 through p-5; px-4 for horizontal; py-4 for vertical.

2. Detailed Class-by-Class Notes

1. container-fluid

Category	Layout
Use Case	Full-width responsive container.
Example	<code><div class="container-fluid">Content</div></code>
Other Options	container — centered max-width container.

2. py-3

Category	Spacing
Use Case	Adds padding to the top and bottom.
Example	<code><div class="py-3">Header</div></code>
Other Options	py-1, py-2, py-4, py-5; p-3 adds padding on all sides.

3. row

Category	Grid
Use Case	Creates a Bootstrap grid row and flex container for columns.
Example	<code><div class="row">...</div></code>
Other Options	row-cols-2, row-cols-3, row-cols-4 can control automatic column counts.

4. align-items-center

Category	Flexbox
Use Case	Vertically centers flex items on the cross axis.
Example	<code><div class="row align-items-center">...</div></code>
Other Options	align-items-start, align-items-end, align-items-stretch.

5. col-md-6

Category	Grid
Use Case	Uses 6 of 12 columns from the md breakpoint upward.
Example	<code><div class="col-md-6">Half</div></code>
Other Options	col-md-3 = 25%; col-md-4 = 33.33%; col-md-8 = 66.67%; col-md-12 = 100%.

6. text-md-end

Category	Text / Alignment
Use Case	Aligns text to the end from md and above.
Example	<code><div class="text-md-end">...</div></code>
Other Options	text-start, text-center, text-end; responsive versions can use sm, lg, xl, xxl.

7. me-3

Category	Spacing
Use Case	Adds margin to the end side; normally right margin in LTR layouts.
Example	<code>Home</code>
Other Options	me-1, me-2, me-4, me-5; ms-3 applies margin to the start side.

8. h-100

Category	Sizing
Use Case	Sets height to 100% of the parent when the parent has a usable height.
Example	<code><div class="h-100">...</div></code>
Other Options	h-25, h-50, h-75, h-auto.

9. align-items-start

Category	Flexbox
Use Case	Aligns flex items to the start of the cross axis; in a row this is normally the top.
Example	<code><div class="row align-items-start">...</div></code>
Other Options	align-items-center, align-items-end, align-items-stretch.

10. col-md-5

Category	Grid
Use Case	Uses 5 of 12 columns from md upward.
Example	<code><div class="col-md-5">...</div></code>
Other Options	col-md-4, col-md-6, col-md-7, col-md-8, etc.

11. col-md-7

Category	Grid
Use Case	Uses 7 of 12 columns from md upward.
Example	<code><div class="col-md-7">...</div></code>
Other Options	col-md-5 pairs with it to make 12 columns.

12. pt-5

Category	Spacing
Use Case	Adds large padding to the top.
Example	<code><div class="pt-5">Hero content</div></code>
Other Options	pt-1 through pt-5; py-5 adds top and bottom padding.

13. container

Category	Layout
Use Case	Responsive centered container with breakpoint-based max-width.
Example	<code><div class="container">Programs</div></code>
Other Options	container-fluid; container-sm, md, lg, xl, xxl.

14. mt-4

Category	Spacing
Use Case	Adds margin to the top.
Example	<code><div class="mt-4">Programs</div></code>
Other Options	mt-1, mt-2, mt-3, mt-5; mb-4 for bottom margin; my-4 for top and bottom.

15. text-center

Category	Text / Alignment
Use Case	Centers inline text and inline content.
Example	<code><div class="text-center">Program</div></code>
Other Options	text-start, text-end.

16. g-3

Category	Grid / Gutters
Use Case	Adds horizontal and vertical grid gutters.
Example	<code><div class="row g-3">...</div></code>
Other Options	g-0 through g-5; gx-3 horizontal only; gy-3 vertical only.

17. col-md-3

Category	Grid
Use Case	Uses 3 of 12 columns from md upward, giving four equal columns.
Example	<code><div class="col-md-3">Program 1</div></code>
Other Options	col-md-6 gives two columns; col-md-4 gives three; col-md-12 gives one.

18. border

Category	Borders
Use Case	Adds a border around the element.
Example	<code><div class="border">Box</div></code>
Other Options	border-0 removes it; border-top, border-end, border-bottom, border-start.

19. p-4

Category	Spacing
Use Case	Adds padding on all four sides.
Example	<code><div class="p-4">Program 1</div></code>
Other Options	p-0 through p-5; px-4 for horizontal; py-4 for vertical.

3. How the Classes Work Together in Your Code

Header

container-fluid py-3 → row align-items-center → col-md-6 + col-md-6

Navigation

col-md-6 text-md-end → me-3 on each navigation link

Hero layout

container-fluid → row h-100 align-items-start → col-md-5 + col-md-7 → pt-5

Programs section

container mt-4 → row text-center g-3 → four col-md-3 columns

Program boxes

border p-4 → border creates the box and p-4 creates internal spacing

4. Important Concepts

- Bootstrap uses a 12-column grid. col-md-3 is 25%, col-md-6 is 50%, col-md-5 is about 41.67%, and col-md-7 is about 58.33%.
- md means the class applies from the medium breakpoint upward. Bootstrap 5 uses 768px as the md breakpoint.
- Spacing utilities use a scale such as 0 to 5. p-4 means padding level 4; mt-4 means top margin level 4.
- row creates the grid/flex row. col-* classes define column widths inside that row.
- g-3 controls the gap between grid rows and columns.
- Responsive classes such as text-md-end allow different behavior at different screen sizes.
- hero-section and logo are NOT Bootstrap classes. They are custom classes defined in the internal style tag.